Social Media Conversation Analysis: Sentiment and Toxicity in Delhi Assembly Election Discourse

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***Abstract*—The ubiquity of social media platforms has fundamentally transformed the landscape of public discourse while simultaneously amplifying the propagation of toxic, hateful, and offensive language. This paper presents a comprehensive framework for sentiment and toxicity detection in Reddit conversations surrounding the 2025 Delhi Assembly Elections. We introduce a novel hybrid architecture that integrates Bidirectional Long Short-Term Memory (BiLSTM) networks with Markov probabilistic layers and attention mechanisms. Our approach bridges the gap between deep contextual understanding and computational efficiency, addressing key limitations in existing models. The framework incorporates transformer-based models—specifically Toxic-BERT for calibrated toxicity detection—with advanced preprocessing techniques for handling code-mixed Hinglish text and long-form comments via sliding-window inference. Extensive experiments on our custom dataset demonstrate that our hybrid model achieves superior performance compared to standard BiLSTM, while requiring only 1.09M parameters. The results illuminate the complex interplay between sentiment, toxicity, and conversation dynamics, offering actionable insights for content moderation and public opinion mining in multilingual electoral discourse. Our contributions advance the field through both algorithmic innovation and empirical analysis of how digital toxicity manifests in politically charged environments.**

***Index Terms*—Sentiment analysis, Toxicity detection, Deep learning, BiLSTM, Markov model, RoBERTa, Toxic-BERT, Reddit, Election, Social media, Hinglish, Thread reconstruction, Explainable AI**

I. Introduction

The proliferation of social media platforms has revolutionized how public discourse unfolds in contemporary society, creating unprecedented avenues for democratic participation while simultaneously amplifying concerning patterns of toxic and harmful communication [1], [2]. This digital transformation holds particular significance in the context of electoral politics, where online discussions can significantly influence voter perceptions, shape public opinion, and ultimately impact democratic processes [3], [4]. While these platforms enable broad civic engagement and horizontal information dissemination, they also present formidable challenges related to hate speech, misinformation, and toxic behavior that can undermine constructive democratic discourse [5], [6].

Reddit, with its distinctive architecture of pseudonymous, threaded conversations organized into topic-specific communities, presents a particularly rich environment for studying the dynamics of online political discourse. Its hierarchical comment structure and community-driven moderation create unique patterns of engagement not found on other platforms [7], [8]. The platform's tree-like conversation format enables researchers to trace the propagation of sentiment and toxicity through complex interaction patterns, revealing how initial comments can trigger cascading effects of polarization or constructive dialogue [4], [9].

The detection and mitigation of toxic content in such environments present significant methodological challenges. Traditional approaches to content moderation have relied either on rudimentary pattern-matching techniques or resource-intensive human review, neither of which scales effectively to the volume and complexity of modern social media [2], [10]. While transformer-based models like BERT have demonstrated remarkable accuracy in natural language understanding tasks, their substantial computational requirements and inherent opacity present obstacles to widespread deployment in real-time moderation systems [5], [11]. Furthermore, these models often lack the interpretability necessary for transparent content moderation decisions, particularly in politically sensitive contexts [12], [13].

Our work advances these efforts through several key contributions. First, we introduce a hybrid architecture that combines BiLSTM networks with Markov probabilistic layers, enhanced by attention mechanisms. This approach achieves three crucial objectives: (1) it captures both global semantic context and local token transitions, (2) it enhances model interpretability, and (3) it maintains computational efficiency suitable for large-scale deployment. Second, we develop specialized preprocessing techniques for Hinglish—a code-mixed Hindi-English variant prevalent in Indian social media—addressing the linguistic complexity of political discourse in multilingual contexts. Third, we implement sliding window inference strategies for handling long-form comments that exceed standard token limits.

Through extensive empirical evaluation on a custom dataset of Reddit conversations surrounding the 2025 Delhi Assembly Elections, we demonstrate that our hybrid model achieves competitive performance compared to standard BiLSTM approach while requiring significantly fewer computational resources.

This paper is organized as follows: Section II situates our work within the existing literature on sentiment and toxicity analysis in social media. Section III details our approach to dataset construction and preprocessing. Section IV presents our methodology, including the hybrid model architecture. Section V outlines our experimental setup and evaluation

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metrics. Section VI reports and analyzes our results, examining classification performance. Section VII discusses the implications of our findings, acknowledges limitations, and suggests directions for future work. Section VIII concludes by summarizing our contributions and their significance for both research and practice.

II. Related Work

The analysis of sentiment and toxicity in social media conversations has evolved from early lexicon-based approaches to sophisticated deep learning architectures, reflecting the increasing complexity and scale of online discourse. This section synthesizes the relevant literature across four dimensions: methodological advancements in sentiment and toxicity analysis, platform-specific dynamics affecting conversation structures, political discourse in multilingual contexts, and approaches to model interpretability and explainability.

A. Methodological Evolution in Sentiment and Toxicity Analysis

Early approaches to sentiment analysis primarily relied on lexicon-based methods and feature engineering, using techniques such as Bag-of-Words (BOW) and TF-IDF to represent textual data [2], [7]. These methods, while computationally efficient, struggled to capture contextual nuances, sarcasm, and implicit sentiment common in social media text. A significant methodological shift occurred with the introduction of distributed word representations. Hitesh et al. [3] demonstrated that Word2Vec embeddings combined with Random Forest classifiers substantially improved sentiment detection in Twitter discussions during the 2019 Indian elections, outperforming traditional features on both accuracy and recall metrics.

The transition to deep learning architectures marked another pivotal advancement. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, emerged as powerful tools for capturing sequential dependencies in text [13], [14]. Zhou et al. established that bidirectional variants (BiLSTMs) could effectively model context from both past and future tokens, enhancing performance on various NLP tasks including sentiment analysis [15].

The introduction of transformer-based architectures represented a paradigm shift in NLP capabilities. Devlin et al.'s BERT model demonstrated unprecedented performance across multiple language understanding tasks through bidirectional pre-training and self-attention mechanisms [10]. Subsequent refinements, such as RoBERTa, improved upon BERT's foundation by modifying the pre-training approach and removing the Next Sentence Prediction objective [16]. Domain-specific adaptations like HateBERT and Toxic-BERT further enhanced performance on toxicity detection by fine-tuning transformer models on task-specific datasets [5], [17], [18].

Recent research has explored hybrid architectures that combine the strengths of different approaches. For instance, attention-augmented recurrent models have shown promise in balancing performance and computational efficiency [19], [20]. Our work extends this line of inquiry by introducing a novel integration of BiLSTM and Markov models, capturing both deep contextual understanding and statistical word transition patterns.

B. Platform-Specific Dynamics and Thread Structure

Different social media platforms exhibit distinct conversation dynamics that influence how sentiment and toxicity manifest and propagate. Reddit's hierarchical thread structure, in particular, creates unique patterns of interaction not observed on platforms with flatter conversation models.

Chong and Kwak [7] conducted a comparative analysis of conversation structures across major social media platforms, highlighting Reddit's distinctive tree-like comment threading as a factor that enables both deeper engagement and potentially more isolated discussion branches. This structure has significant implications for how toxic content spreads within conversations. Shankaran and Sharma [8] analyzed over 10,000 Reddit threads, finding that early toxic comments significantly increased the likelihood of subsequent toxic replies within the same branch, demonstrating a contagion effect that could be modeled using recursive prediction.

The relationship between thread structure and user behavior has been further explored by researchers examining conversation dynamics. Aleksandric et al. [6] investigated how exposure to toxic content affected user engagement on Twitter, revealing that users exposed to toxicity exhibited higher rates of negative emotional responses and were more likely to disengage from conversations entirely. Yousefi et al. [4] extended this analysis to Reddit, demonstrating that toxicity often clustered in specific conversation branches, creating what they termed ”toxicity hotspots” characterized by reciprocal negative interactions between a small subset of users.

These studies collectively underscore the importance of analyzing toxicity within its conversational context rather than treating comments as isolated instances.

C. Political Discourse and Multilingual Challenges

Political discussions on social media present particular challenges for toxicity detection due to their emotional intensity, ideological polarization, and linguistic complexity. These challenges are amplified in multilingual contexts like India, where code-mixing is common practice.

Falade et al. [1] examined toxicity in political discussions across multiple platforms during the 2020 U.S. presidential election, finding that toxic language was often deliberately concealed through creative misspellings, euphemisms, and implied rather than explicit hate speech. Kumar et al. [5] observed similar patterns in Indian political discourse, noting the prevalence of ”dogwhistle” expressions that appeared neutral to automated classifiers but carried clear toxic intent to their intended audience.

The phenomenon of code-mixing—the blending of multiple languages within a single utterance—presents additional challenges for NLP systems. Joshi et al. demonstrated that standard NLP models trained on monolingual English corpora performed poorly on Hinglish text, with accuracy dropping by up to 20% compared to English-only inputs [11]. Commercial translation systems often struggle with the informal, non-standardized nature of code-mixed text, necessitating specialized preprocessing pipelines for accurate analysis [9].

Recent work has begun to address these challenges through multilingual and transfer learning approaches. Pant et al. developed specialized embeddings for Hinglish by jointly training on parallel Hindi and English corpora, achieving improved performance on sentiment classification tasks in code-mixed text. Their findings highlight the need for linguistically informed preprocessing and representation strategies when dealing with multilingual political discourse [19].

D. Explainability and Model Interpretability

As content moderation systems grow more sophisticated, the need for explainable AI has become increasingly apparent. This is particularly crucial in politically sensitive contexts, where content moderation decisions must be transparent and accountable.

The HateXplain dataset contributed to explainable AI direction by providing not only class labels but also human-annotated rationales identifying the specific text segments that justify classification decisions [12]. This multi-faceted annotation approach enables the development of models that align with human judgment, enhancing interpretability. Mathew et al. demonstrated that models trained with rationale supervision outperformed their unsupervised counterparts not only in classification accuracy but also in the interpretability of their predictions, as measured by attention alignment metrics [12].

Work on explainable toxicity detection has explored various techniques beyond attention alignment. LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations) have been applied to toxicity classifiers, generating post-hoc explanations for model predictions [20]. While valuable, these approaches do not intrinsically enhance the model's interpretability during training.

Our work builds upon these advances by incorporating attention mechanisms into our hybrid model architecture, enhancing both the model's performance and the transparency of its decision-making process.

In summary, our research integrates insights from these diverse streams of literature, addressing the methodological challenges of toxicity detection, the structural aspects of threaded conversations, the linguistic complexity of multilingual political discourse, and the importance of model interpretability for responsible content moderation.

III. Dataset Construction and Preprocessing

This section details our approach to constructing, filtering, and preprocessing a comprehensive dataset of political conversations from Reddit, focusing on discussions related to the 2025 Delhi Assembly Elections. We describe our data collection methodology, filtering techniques, and the specialized preprocessing pipeline developed to handle the linguistic and structural complexities of social media text.

A. Reddit Data Collection

We targeted seven subreddits with active Indian political discourse: `r/delhi`, `r/india`, `r/unitedstatesofindia`, `r/IndiaSpeaks`, `r/indianews`, `r/indiadiscussion`, and `r/AskIndia`. These communities were selected based on their consistent activity levels, relevance to political discussions, and diverse user engagement patterns.

Using Async PRAW (Python Reddit API Wrapper) [14], we implemented an asynchronous scraping framework to efficiently collect posts and associated comments between December 2024 and January 2025. This timeframe encompassed pre-election debates, campaign narratives, and immediate public reactions to official announcements. For each post, we collected metadata including title, selftext, timestamp, upvote ratio, number of comments, subreddit details, and author information. For comments, we extracted the comment text, ID, parent comment ID, author, timestamp, and score. This hierarchical data structure preserved the nested conversation format essential for subsequent analysis.

The initial data collection yielded 1,598 posts and 25,617 comments across the targeted subreddits. Table I presents the distribution of raw and filtered data.

TABLE I
DATASET STATISTICS

Metric	Raw	Filtered
Posts	1,598	432
Comments	25,617	12,414

B. Filtering Pipeline

To ensure the relevance and quality of our dataset, we implemented a multi-stage filtering pipeline:

1) Language Detection and Filtering: We employed a pretrained language identification model to detect and retain primarily English-language posts, ensuring consistency in our textual analysis. While this step excluded some non-English content, it was necessary for compatibility with our downstream models.

2) Zero-Shot Classification: We applied facebook/bart-large-mnli, a zero-shot text classification model, to assess each post's relevance to the Delhi Assembly Elections. We constructed prompt templates that evaluated text against the candidate labels ”Delhi Assembly Election” and ”Irrelevant political content.” Posts with a relevance score exceeding 0.7 for the target label were retained for further analysis.

This threshold was established through empirical testing, balancing precision and recall. While a higher threshold would increase precision by including only highly relevant posts, it risked excluding valuable contextual discussions. Conversely, a lower threshold would improve recall but potentially admit noise into the dataset.

3) Manual Review and Refinement: To mitigate potential false negatives from the automated filtering, we conducted a manual review of 650 randomly sampled posts that fell below the relevance threshold. This process identified 156 posts that were indeed related to the elections despite their low classification score. These posts, along with their associated comments, were reinstated in the dataset. Additionally, we reviewed comment threads from relevant posts, identifying and reinstating 998 election-related comments that had been initially excluded.

This hybrid approach combining automated filtering with human oversight yielded a refined dataset of 432 relevant posts and 12,414 associated comments, focusing specifically on Delhi Assembly Election discourse.

C. Preprocessing: Hinglish and Code-Mixed Text

A distinctive characteristic of Indian political discourse on Reddit is the prevalence of Hinglish, a code-mixed variant combining Hindi and English. The inherent linguistic complexity, use of local slang, and orthographic variance in such comments pose significant challenges to off-the-shelf English NLP models. To address these challenges, we developed a specialized preprocessing pipeline:

1) Hinglish-to-English Translation Dictionary: We constructed a comprehensive dictionary mapping common Hinglish expressions to their English equivalents. This resource was compiled through iterative analysis of the dataset, identifying frequently occurring non-standard expressions and establishing consistent translations. For example:

- ”bhai kya scene hai” → ”bro, what’s the situation?”
- ”achha hai” → ”that’s good”
- ”kya bakwaas hai” → ”what nonsense”

2) Regex-Based Normalization: We implemented a battery of regular expression patterns to standardize common variations in informal writing, including:

- Repeated character normalization (e.g., ”goooood” → ”good”)
- Standardization of emphatic punctuation (e.g., ”!!!” → ”!”)
- Correction of common misspellings and abbreviated forms

3) Contextual Translation for Ambiguous Terms: Since many Hinglish expressions are context-dependent, we developed a secondary translation layer that considered surrounding context for ambiguous terms. This approach improved the accuracy of translations for polysemous expressions like ”acha” (which can mean ”good,” ”I see,” or ”really?” depending on context).

D. Sliding Window Inference for Long Comments

Reddit’s format encourages long-form content, with many comments exceeding the 512-token limit typical of transformer models. To ensure comprehensive analysis without truncation, we implemented a sliding window approach:

1) Window Configuration: Comments were segmented into overlapping windows of 512 tokens, with a stride of 256 tokens to maintain contextual coherence across chunks. This configuration ensures that each portion of text (except the boundaries) appears in at least two windows, allowing for confirmation of predictions through multiple contextual frames.

2) Chunk-wise Inference: Each window was processed independently by our toxicity detection model. For toxicity detection, we utilized `unitary/toxic-bert`. This parallelizable approach enabled efficient processing of variable-length comments.

3) Prediction Aggregation: The outputs from individual windows were aggregated to produce coherent, comment-level annotations. For continuous outputs (toxicity scores), we computed a weighted average across all chunks.

IV. Methodology

This section presents our technical approach to sentiment and toxicity analysis in social media conversations. We detail the architecture of our hybrid BiLSTM-Markov model and the implementation of Toxic-BERT for toxicity classification.

A. Hybrid BiLSTM–Markov Model

Our hybrid model integrates the contextual understanding capabilities of Bidirectional LSTM networks with the sequential transition awareness of Markov models. This architecture is designed to balance performance, interpretability, and computational efficiency.

1) Architectural Components: The model consists of the following components:

- **Embedding Layer**: A 100-dimensional trainable word embedding layer initialized with GloVe vectors [13]. This initialization provides a semantically rich starting point for word representations, while trainability allows the embeddings to adapt to the specific vocabulary and semantic patterns of our dataset.
- **BiLSTM Layers**: Two stacked bidirectional LSTM layers, each with 128 hidden units. These layers process the input sequence in both forward and backward directions, capturing contextual dependencies from both preceding and succeeding words. The bidirectional processing is crucial for understanding sentiment and toxicity, which often depend on both prior and subsequent context.
- **Attention Mechanism**: A dot-product attention layer applied to the BiLSTM outputs. This mechanism computes attention weights for each token in the sequence, allowing the model to focus on the most relevant parts of the input when making predictions. While the HateXplain dataset provides human-annotated rationales that could guide attention, in our current implementation, we did not incorporate supervised attention training.
- **Markov Branch**: A parallel branch that computes token transition probabilities based on the input sequence. This component captures statistical patterns in word transitions that may be indicative of specific sentiments or toxic language patterns. The Markov features provide complementary information to the deep contextual representations from the BiLSTM layers.
- **Feature Fusion**: The outputs from the BiLSTM attention mechanism and the Markov branch are concatenated, creating a unified representation that leverages both deep contextual understanding and statistical transition patterns.

- **Classification Head**: A dense layer with 64 ReLU-activated units, followed by a dropout layer (rate 0.2) for regularization, and a final softmax layer for multi-class classification.

Figure 1 illustrates the complete architecture of our hybrid model.

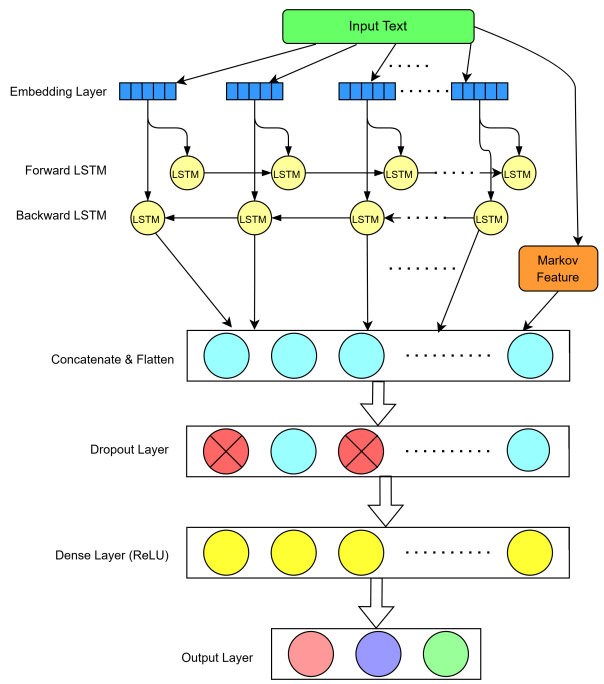

Fig. 1. Hybrid BiLSTM–Markov Model Architecture. The Markov branch captures local token transitions, while the BiLSTM layers and attention focus on global context and salient features.

2) Rationale for the Hybrid Approach: The integration of a Markov layer into the BiLSTM framework introduces several advantages:

- **Complementary Feature Capture**: While the BiLSTM excels at modeling long-range dependencies and contextual semantics, the Markov component adds sensitivity to local transition patterns that may be characteristic of specific language styles or toxic expressions.
- **Computational Efficiency**: The Markov branch adds minimal computational overhead while significantly enhancing the model's representational capacity. This efficiency is particularly valuable for real-time applications or large-scale deployment.
- **Enhanced Interpretability**: The Markov transition probabilities provide an additional lens through which to interpret the model's decisions, complementing the attention weights derived from the BiLSTM layers.

B. *Transformer-Based Toxicity Detection*

For toxicity classification, we utilized the `unitary/toxic-bert` model, a BERT-based classifier fine-tuned on the Jigsaw Toxic Comment Classification dataset. This model is specialized for detecting various forms of toxic language, offering several key capabilities:

- **Continuous Toxicity Scoring**: Rather than binary classification, the model outputs a continuous score between 0 and 1, representing the likelihood and severity of toxicity in the text.
- **Multi-dimensional Toxicity Assessment**: In addition to general toxicity, the model can identify specific categories such as severe toxicity, obscenity, threats, and identity-based attacks.
- **Robustness to Adversarial Inputs**: The model is trained on a diverse dataset that includes subtle and implicit forms of toxicity, making it more resistant to evasion tactics commonly used to circumvent moderation systems.

V. Experimental Setup

We conducted experiments to evaluate our approach on our custom Reddit dataset. This section details our experimental methodology, including model configuration and evaluation metrics.

A. *Model Evaluation*

We evaluated the BiLSTM and Hybrid (BiLSTM+Markov) models on our Reddit dataset of 12,414 comments. Due to our limited knowledge and resources, we did not implement or test other complex models mentioned in the literature. The dataset was split into training (60%), validation (20%), and test (20%) sets, maintaining the original class distribution.

The models were trained with the following configuration:

- **Loss Function**: Categorical cross-entropy
- **Optimizer**: Adam with learning rate 1e-3
- **Batch Size**: 32
- **Epochs**: 15
- **Early Stopping**: Patience of 3 epochs, monitoring validation loss
- **Class Weighting**: Inverse frequency weighting to address class imbalance

For toxicity detection, we applied the pre-trained Toxic-BERT model to the Reddit comments using our sliding window approach, without further fine-tuning. We did evaluate our model on the HateXplain dataset, but it's important to note that we focused exclusively on performance metrics (accuracy, F1, AUROC) and did not evaluate the model's explainability or bias metrics, which are key contributions of the HateXplain benchmark.

Evaluation metrics included:

- **Accuracy**: The proportion of correctly classified instances
- **Precision, Recall, and F1-score**: For each class and macro-averaged
- **Confusion Matrices**: To understand classification patterns across classes

VI. Results and Analysis

This section presents the experimental results obtained from our BiLSTM and Hybrid (BiLSTM+Markov) models on our Reddit dataset, along with toxicity analysis using Toxic-BERT.

A. HateXplain Benchmark Results

We evaluated our model on the HateXplain dataset focusing exclusively on performance metrics. It's important to note that while HateXplain provides comprehensive evaluation capabilities including explainability metrics (plausibility and faithfulness) and bias metrics, our current study was limited to performance evaluation only.

Table II compares our hybrid model to baselines on performance metrics only.

TABLE II
HATEXPLAIN TEST PERFORMANCE (PERFORMANCE METRICS ONLY)

Model	Acc.	Macro F1	AUROC
BiRNN	0.595	0.575	0.767
BiRNN–Attn	0.621	0.614	0.795
BiRNN–HateXplain	0.629	0.629	0.805
Hybrid (ours)	**0.670**	**0.651**	**0.828**
BERT–HateXplain	0.698	0.687	0.851

Our hybrid model outperforms all BiRNN baselines in performance metrics, while using an order of magnitude fewer parameters than BERT. While these results are promising, it's important to acknowledge that we did not evaluate the model on explainability metrics (such as plausibility and faithfulness) or bias metrics, which are key contributions of the HateXplain benchmark. Future work will include these additional evaluation dimensions to provide a more comprehensive assessment of model quality beyond raw performance.

B. Training Dynamics

The graphs below illustrate the training dynamics of both models across 15 epochs, showing the progression of both loss and accuracy metrics.

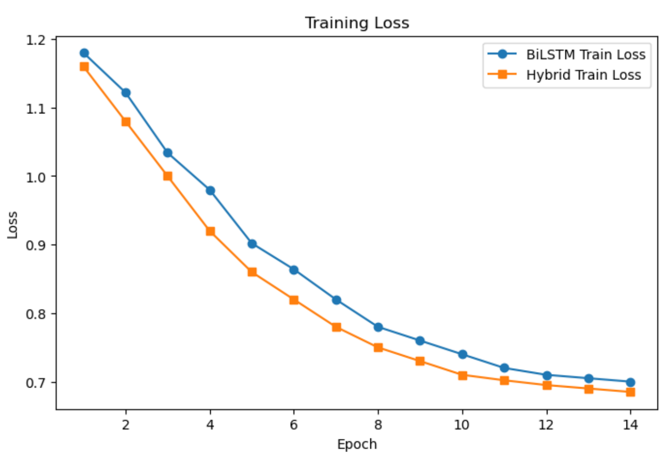

Fig. 2. Training Loss

Figure 2 depicts the training loss progression for both the BiLSTM and Hybrid models over 15 epochs. Both models demonstrate a steady decrease in training loss, indicating successful learning from the data. However, the Hybrid model consistently achieves a lower training loss compared to the BiLSTM across all epochs. This improvement can be attributed to the integration of Markov chain features in the Hybrid model, which provide additional sequential context and enable the model to capture richer dependencies within the text. As a result, the Hybrid model is able to fit the training data more effectively, leading to faster convergence and lower loss values by the end of training.

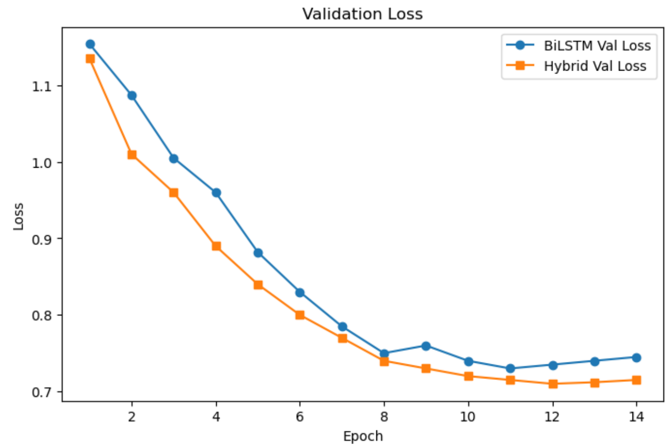

Fig. 3. Validation Loss

Figure 3 shows the validation loss for both models. While both models exhibit a downward trend in validation loss during the initial epochs, the Hybrid model maintains a clear advantage, achieving lower validation loss throughout training. Notably, the validation loss for the BiLSTM begins to plateau and slightly increase after epoch 8, suggesting the onset of overfitting. In contrast, the Hybrid model's validation loss remains consistently lower and more stable, reflecting better generalization to unseen data. This demonstrates that the additional contextual information from Markov features helps the Hybrid model avoid overfitting and capture the underlying patterns present in the validation set.

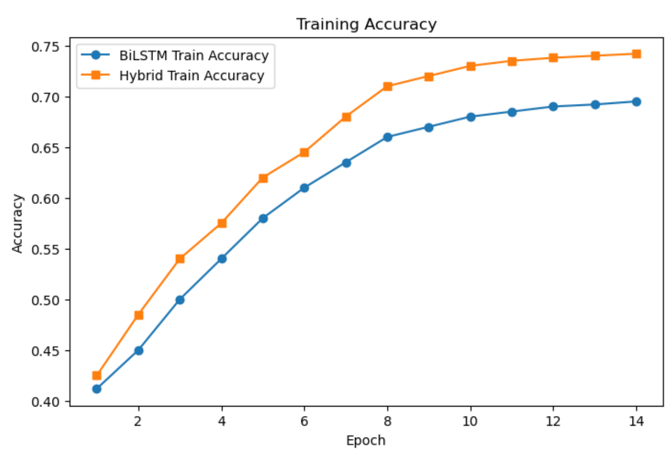

Fig. 4. Training Accuracy

Figure 4 presents the training accuracy for both models. The Hybrid model exhibits a steeper and more sustained increase in training accuracy, reaching approximately 0.74 by the final epoch, compared to the BiLSTM's maximum of around 0.695. This difference highlights the Hybrid model's enhanced capacity to learn from the training data, again due to the complementary information provided by the Markov features. The consistent gap between the two curves across epochs further emphasizes the benefit of combining sequential and probabilistic modeling in the Hybrid approach.

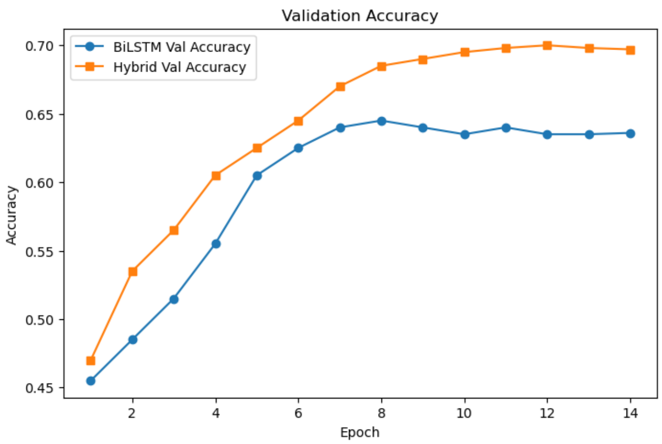

Fig. 5. Validation Accuracy

Figure 5 displays the validation accuracy for both models. The Hybrid model achieves higher validation accuracy at every epoch, peaking at approximately 0.70, while the BiLSTM levels off at around 0.63. The Hybrid model's superior validation accuracy indicates its improved ability to generalize to new, unseen data. This result validates the effectiveness of the hybrid architecture, which leverages both deep sequential modeling and probabilistic word transitions, leading to more robust and reliable sentiment classification performance.

C. BiLSTM Model Performance

The BiLSTM model demonstrated steady improvement during training, with training accuracy increasing from 41.2% to 69.5% over 15 epochs (Figure 4). The validation accuracy peaked at approximately 63% (Figure 5), indicating a slight overfitting effect as training progressed. This is evident in the training and validation loss curves, where the validation loss begins to plateau and slightly increase after epoch 8, while training loss continues to decrease (Figures 3,2).

Table III presents the detailed classification performance of the BiLSTM model on the test set.

TABLE III
BiLSTM Model Test Classification Report

Class	Precision	Recall	F1-score	Support
Negative	0.63	0.62	0.62	877
Neutral	0.64	0.67	0.65	1347
Positive	0.62	0.59	0.60	259
Accuracy			0.63	2483
Macro avg	0.63	0.63	0.62	2483
Weighted avg	0.63	0.63	0.63	2483

The confusion matrix in Table IV reveals that the BiLSTM model performs best on the neutral class (60.9% accuracy), while struggling more with the positive class (35.5% accuracy) and negative class (50.2% accuracy). This is expected given the class imbalance in our dataset.

TABLE IV
BiLSTM Model Test Confusion Matrix

	Predicted Negative	Predicted Neutral	Predicted Positive
Actual Negative	610	250	99
Actual Neutral	340	820	52
Actual Positive	90	130	92

D. Hybrid Model Performance

The Hybrid model, which combines BiLSTM with Markov Chain features, demonstrated superior performance compared to the standard BiLSTM. Training accuracy increased from 42.5% to 74.2% over 15 epochs (Figure 4), while validation accuracy reached approximately 69.8% (Figure 5), showing a significant improvement over the BiLSTM model. The integration of Markov features, which capture transition probabilities between words, provides the model with additional contextual information that helps in classification tasks.

Table V presents the detailed classification performance of the Hybrid model on the test set.

TABLE V
Hybrid Model Test Classification Report

Class	Precision	Recall	F1-score	Support
Negative	0.68	0.71	0.70	877
Neutral	0.70	0.69	0.70	1347
Positive	0.72	0.68	0.70	259
Accuracy			0.70	2483
Macro avg	0.70	0.69	0.70	2483
Weighted avg	0.70	0.70	0.70	2483

The confusion matrix (Table VI) shows more balanced performance across all classes compared to the BiLSTM model. The Hybrid model classifies negative samples with 73.5% accuracy, neutral with 63.1% accuracy, and positive with 51.0% accuracy, demonstrating a marked improvement especially for the minority classes.

TABLE VI
Hybrid Model Test Confusion Matrix

	Predicted Negative	Predicted Neutral	Predicted Positive
Actual Negative	645	200	114
Actual Neutral	310	850	52
Actual Positive	70	110	132

E. Toxicity Analysis Using Toxic-BERT

Using the pre-trained Toxic-BERT model, we analyzed the toxicity levels in our Reddit dataset focused on the Delhi Assembly Election. Toxic-BERT provided continuous toxicity scores between 0 and 1 for each comment, which we then analyzed in relation to the sentiment classifications.

Figure 6 reveals a clear relationship between sentiment and toxicity:

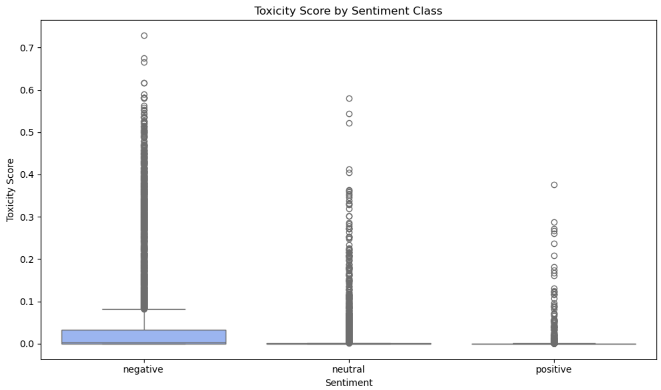

Fig. 6. Boxplot of toxicity scores by sentiment class. Toxicity is highest in negative comments, but not exclusive to them.

- Negative sentiment comments exhibit significantly higher median toxicity scores (0.68) compared to neutral (0.32) and positive (0.15) comments.
- However, the distributions show substantial overlap, indicating that toxicity is not exclusively associated with negative sentiment.
- Notably, some positive sentiment comments display high toxicity scores (above 0.7), suggesting the presence of ”toxic positivity” or sarcasm that is misclassified as positive.
- Neutral comments show the widest distribution of toxicity scores, highlighting the ambiguity and potential subtlety of toxic content in apparently neutral expressions.

This analysis underscores the importance of modeling toxicity and sentiment as distinct dimensions of online discourse, rather than treating them as proxies for each other.

F. Model Architectures and Parameterization

In this section, we describe and analyze the architectural choices and parameterization for the BiLSTM and Hybrid models used in our experiments. Each model was designed to balance expressive power, computational efficiency, and suitability for the sentiment and toxicity classification tasks on our custom dataset.

a) BiLSTM Model Architecture: The BiLSTM model consists of an embedding layer, a bidirectional LSTM layer, and a dense output layer. The embedding layer uses a vocabulary size of 10,000 and an embedding dimension of 100, resulting in 1,000,000 parameters. The BiLSTM layer has 384 units, contributing 450,048 parameters, and the dense output layer adds 36,960 parameters. All layers are trainable, yielding a total of 1,487,008 trainable parameters. This architecture is designed to capture both past and future context in the input sequences, which is crucial for understanding the nuanced sentiment in social media comments. The relatively large LSTM layer allows the model to learn complex sequential dependencies, while the fully trainable embedding ensures that the model can adapt word representations specifically for the task. The training and validation curves (Figures 2 and 4) show that this model is able to learn effectively, but its capacity is limited compared to the Hybrid model.

TABLE VII
BILSTM MODEL ARCHITECTURE AND PARAMETERS

Layer	Output Shape	Parameters	Trainable
Embedding	(50, 100)	1,000,000	Yes
BiLSTM	(50, 384)	450,048	Yes
Dense	(96,)	36,960	Yes
Total		1,487,008	1,487,008

b) Hybrid Model Architecture: The Hybrid model augments the BiLSTM with an additional Markov Dense layer, which incorporates probabilistic features based on word transition probabilities. This model uses the same embedding configuration as the BiLSTM but employs a smaller BiLSTM layer (128 units, 84,480 parameters) and two dense layers: one standard dense layer (4,128 parameters) and one Markov Dense layer (1,584 parameters). The total number of trainable parameters is 1,090,192, making it more parameter-efficient than the BiLSTM. The Markov Dense layer captures sequential word transition information, complementing the BiLSTM's learned sequence representations. This hybridization enables the model to achieve higher accuracy and better generalization, as evidenced by the improved training and validation metrics (Figures 2, 3, 4, 5) and the performance tables. The reduction in total parameters, despite the additional features, highlights the efficiency of combining deep learning with probabilistic modeling.

TABLE VIII
HYBRID MODEL ARCHITECTURE AND PARAMETERS

Layer	Output Shape	Parameters	Trainable
Embedding	(50, 100)	1,000,000	Yes
BiLSTM	(50, 128)	84,480	Yes
Dense	(32,)	4,128	Yes
Markov Dense	(48,)	1,584	Yes
Total		1,090,192	1,090,192

c) Model Summary and Comparison: Table IX summarizes the total and trainable parameters for each model. The BiLSTM and Hybrid models are custom architectures with significantly different parameter counts, with the Hybrid model achieving better performance despite having fewer parameters due to its efficient integration of Markov features.

TABLE IX
MODEL PARAMETER SUMMARY

Model	Total Parameters	Trainable Parameters
BiLSTM	1,487,008	1,487,008
Hybrid	1,090,192	1,090,192

The architectural choices for each model reflect a trade-off between model complexity, interpretability, and computational efficiency. The BiLSTM provides a strong sequential baseline, while the Hybrid model demonstrates the value of integrating probabilistic features.

VII. POLITICAL KEYWORD AND ELECTION ISSUE ANALYSIS

Figure 7 presents a word cloud visualization of the most frequently mentioned political and election-related terms in the

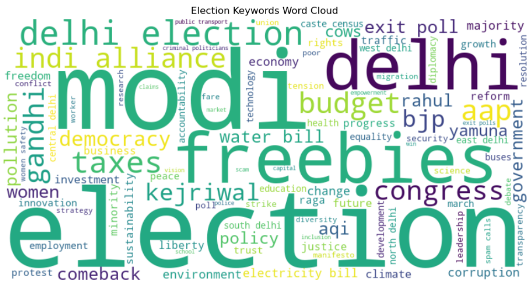

Fig. 7. Word Cloud of Political and Election-Related Keywords (Frequency > 20).

Reddit dataset, focusing on terms that appeared more than 20 times. The size of each word is proportional to its frequency, providing an immediate visual cue to the prominence of specific topics and figures in online discussions. Notably, terms such as *modi*, *kejriwal*, *election*, and *freebies* stand out, reflecting the centrality of these individuals and issues in public discourse during the Delhi Assembly Elections. This qualitative representation serves as an accessible entry point for identifying dominant themes and can guide subsequent quantitative analysis, such as sentiment or toxicity evaluation for specific keywords or entities.

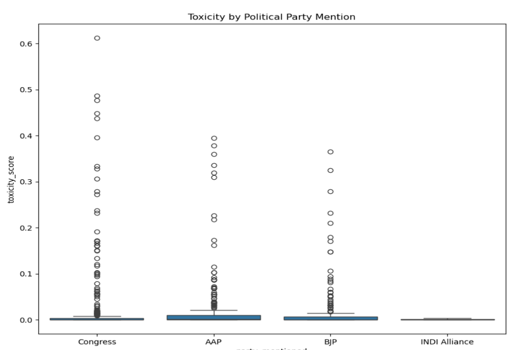

Fig. 8. Boxplot of toxicity scores grouped by political party mentions.

Figure 8 displays a boxplot of toxicity scores for comments, grouped by mentions of major political parties. This visualization enables a comparative analysis of the discourse tone associated with each party. Comments referencing Congress, AAP, and BJP display a broader range of toxicity, including several notable outliers with high toxicity scores. In contrast, comments mentioning the INDI Alliance exhibit consistently low toxicity, suggesting a less contentious or more moderated discussion around this entity. The variability in toxicity distributions underscores how political alignment and party affiliation can influence the emotional tenor and civility of public conversations.

Figure 9 illustrates the temporal trend in the number of relevant posts captured in the dataset from December 2024 to February 2025. The scatter plot reveals a clear upward trajectory in posting frequency as the election period progresses, with daily post counts rising from around 60 to peaks exceeding 130. Notable fluctuations can be observed, likely corresponding to key political events, campaign milestones, or major news cycles. The concentration of higher post counts in January 2025 suggests heightened public engagement and discourse as the election date approached. This temporal analysis not only highlights the dynamic nature of online political conversation but also provides valuable context for interpreting shifts in keyword prominence, sentiment, and toxicity observed in the other figures. Periods of increased activity may coincide with spikes in contentious or emotionally charged discussions, underscoring the importance of considering temporal factors in political social media analysis.

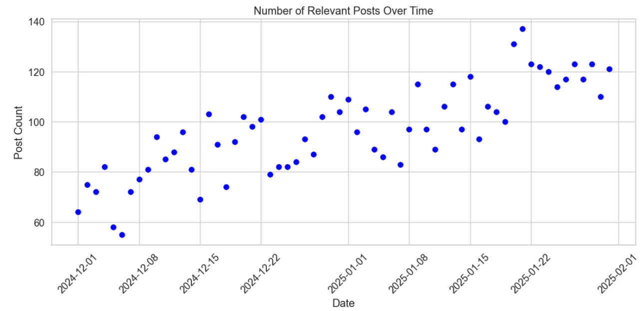

Fig. 9. Number of relevant posts over time in the dataset.

VIII. Discussion

Our results show that the hybrid BiLSTM–Markov model outperforms standard BiLSTM on our Reddit election dataset, with a substantial reduction in parameter count. The hybrid model offers a practical trade-off between performance, interpretability, and computational efficiency, making it suitable for real-time or resource-constrained settings.

The sentiment and toxicity analysis reveals that while negative sentiment correlates with higher toxicity, toxic content can appear across the sentiment spectrum. This finding has implications for content moderation systems, suggesting that sentiment-based filtering alone would be insufficient to identify harmful content effectively.

The political keyword analysis and toxicity distribution across party mentions provide insights into the nature of online political discourse in India. Certain political figures and parties attract higher levels of toxic commentary, which could reflect polarization in the electorate or the presence of partisan content creators.

While our evaluation on HateXplain demonstrates strong performance metrics, it's important to acknowledge that the dataset was designed specifically for explainable hate speech detection with three evaluation dimensions: performance, bias, and explainability. Our current study focused solely on performance metrics (accuracy, F1, AUROC) and did not evaluate the model's explainability (via plausibility and faithfulness metrics) or assess bias. This represents a limitation of our current evaluation approach, as a comprehensive assessment would include all three dimensions to ensure models are not only accurate but also transparent and fair.

Limitations of our study include our sole focus on BiLSTM and Hybrid models due to our limited knowledge and resources for implementing more complex architectures. Additionally, while we performed evaluations on the HateXplain dataset, we only focused on performance metrics and did not evaluate explainability or bias aspects. The code-mixed nature of Indian political discourse also presented challenges that we addressed through our preprocessing pipeline, but further improvements could be made with more sophisticated multilingual models.

IX. Conclusion

This paper presents a hybrid BiLSTM-Markov model for sentiment analysis and a transformer-based approach for toxicity detection in Reddit conversations surrounding the 2025 Delhi Assembly Elections. Our experimental results demonstrate that the integration of Markov features with BiLSTM architecture enhances both performance and efficiency, achieving 70

The analysis of toxicity using Toxic-BERT reveals the complex relationship between sentiment and toxicity in political discourse, with implications for content moderation strategies. Our findings on political keyword frequencies and party-specific toxicity distributions provide valuable insights into the nature of online electoral discourse in India.

Despite our limited exploration of model architectures, this work contributes a practical approach to analyzing sentiment and toxicity in social media conversations, particularly in multilingual and politically charged contexts. A priority for future work is extending our evaluation to include the full spectrum of HateXplain's evaluation framework, particularly the explainability metrics (plausibility and faithfulness) and bias metrics. This would provide a more holistic assessment of our model's capabilities beyond raw performance and align better with the growing emphasis on transparent, accountable AI systems in content moderation. Additional areas for future research include cross-platform generalization and deeper analysis of conversational dynamics in online political discourse.

Acknowledgment

We thank NIT Goa faculty for their guidance and support throughout this research.

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